Blessed G/Michael Catholic Secondary School

2nd Semester Mathematics per- Test one for Grade 12th,

February/2018 E.C Time Allowe : 1:30

Name _____ NO_____ Section ----------

I. Choose the best answer among the given alternatives.

1. Let the relation (R) = {(x,y), x,y∈lR, $y \leq x + 1$ and $y - 2 \geq -x$} the range of the relation is _____

A. $(\infty, -\infty)$ B. $(-\infty, \frac{3}{2}]$ C. $[\frac{3}{2}, \infty)$ D. $(\frac{3}{2}, 4)$

2. Which one of the following is the inverse of the relation. R={(x,y):x,y∈lR and y = 2x- 4}

A. R^{-1}= {(x,y) , x, y∈ lR and y = 2x − 4 }

C. R^{-1}= {(x,y): y = 2x-2}

B. R^{-1} = {(x,y) $y = \frac{1}{2}x + 2$ }

D. R^{-1} = {(x,y) : y= x+2}

3. The domain of $f(x) = \sqrt{-x + |x|}$ is

$-x + |x| \geq 0$

$|x| \geq x$

A. $x \geq 0$ B. lR C. $x \leq 0$ D. $x \geq 1$

4. Which one of the following relations are functions.

A. R: {(x,y) : x,y∈ lR and $y = x + 2$}

C. R= {(6,1)(7,3)(0,2),(6,4)}

B. R= {(x,y):x,y∈ lR and $x^2 + y^2 = 9$}

D. R= {(-1,0)(0,-3),(2,-3),(3,0)}

5. The solution set of sgn(x^2-4) =_____

A. [-2, 2] B. [-2,∞) C. $(-\infty, -2) U (2, \infty)$ D. $[0, \infty)$

6. The solution set of $[x + 1] = 4$ is _____

A. [3,4) B. (3,4) C. (4,∞) D. $(-3, \infty)$

7. Let $f(x) = x^2+1$ and $g(x) = \sqrt{x}$ then the value of (gof)(-1) is _____

A. 4 B. 2 C. $\sqrt{2}$ D. 8

8. The solution set of $\frac{1}{x-2} = \frac{3}{x+2} - \frac{6x}{x^2-4}$ is _____

A. $\{\frac{-1}{2}, \frac{1}{2}\}$ B. $\{2, -4\}$ C. $\{-2\}$ D. ϕ

9. Which one of the following is the decomposition of $\frac{6x+2}{x^2+6x+5}$ as sum of partical fractions.

A. $\frac{7}{x+5} - \frac{1}{x+1}$ B. $\frac{7}{x+5} + \frac{3}{x+1}$ C. $\frac{8}{x+1} + \frac{2}{x+5}$ D. $\frac{x}{x+2} - \frac{1}{x+5}$

10. Which one of the following is not true about $f(x) = \frac{3x^2+1}{x^2+4x+4}$

A. horizontal asymptote y = 3

C. has an oblique asymptote

B. Vertical asymptote to the line x = -2

D. for $x \to \infty, y \to 3$

D 11. Let A and B be invertible matrices of the same size then which of the following is not true.

A. $(A^{-1})^{-1} = A$ B. $(A^t)^{-1} = (A^{-1})^t$ C. $(AB)^{-1} = B^{-1}A^{-1}$ D. $(AB)^t = A^tB^t$

B 12. Multiply a colum of a matrix by anon- zero constant is called.

A. swapping B. scaling C. pivoting D. scalar multiplication of matrix.

A 13. Which one of the following is the inverse of $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$

A. $\begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$ B. $\begin{pmatrix} \frac{1}{2} & 1 \\ 0 & 1 \end{pmatrix}$ C. $\begin{pmatrix} 0 & \frac{1}{2} \\ 0 & 1 \end{pmatrix}$ D. $\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix}$

B 14. Let A= $\begin{pmatrix} 1 & 4 & 5 \\ 0 & 2 & 3 \\ 0 & 0 & 6 \end{pmatrix}$ then what is the determinant of A.

A. 9 B. 12 C. 0 D. 8

A 15. Let A = $\begin{pmatrix} x & 0 & 0 \\ 0 & x & 0 \\ 0 & 0 & x \end{pmatrix}$ and it is determinant is 8. What is the value of x.

A. 2 B. 3 C. 1 D. 0

D 16. Which one of the following is a function?

A. {(x,y), y is a son of x } B. {(x,y): $y^2 = x$ C. {(1,2),(2,2)(1,4)} D. {(x,y): -1<x<1 and

C 17. Which one of the following is false

A. There are functions whose inverses are not functions

B. The domain of f(x) = sgnx is all real number

C. Let $f(x) = x^3 - x + 1$, the inverse is a function

D. if $f(x) = \frac{x+2}{x-2}$, then f(-2) = 0

B 18. Which one of the following is true about f(x) = sgnx

A. domain of f is (-1,1)

B. Range of f is {-1,0,1}

C. symmetric with respect to y- axis

D. sgn x= $|x|$

C 19. Which one of the following is true

A. $\lfloor x \rfloor \geq x$

B. $\lfloor x \rfloor + k = \lfloor x + k \rfloor, for\ k \in integer$

C. $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$

D. *The range of* $\lfloor x \rfloor$ is $\mathbb{R}$

B 20. The solution set of $|x - 3| + 1 = 0$ *is* _____

A. 2 B. ϕ C. 0 D. 1

C 21. Which one of the following is true about $f(x) = x^{\frac{-10}{7}}$

A. Domain is $\mathbb{R}$

B. Range is $y \geq 0$

C. *symmetry with respect to* $y - axis$

D. The ,properties of $f(x0 = x^{\frac{-10}{7}}$ is the same as f(x

22. Let f(x) = 3x-1 and g(x) = 2x+4, then the value of k for (gof)(k) = 4 is

A. 4 B. $\frac{1}{2}$ C. -3 D. $\frac{1}{3}$

23. Which of the following pairs of function are inverse each other?

A. f(x)= $\sqrt{x}, g(x) = x^2$ B. f(x)= $log_3^{2x}, g(x) = \frac{1}{2}(3^x)$

C. f(x)= $\sqrt{x} + 4, f(x) = x^2$ D. f(x) = $\sqrt{x}, g(x) = |x|$

24. Two planes leave an airport flying at the same rate. If the first plane flies 1.5 hours longer than the second Plane and travels 2000 miles while the second plane travels only 1000 miles for how long was the first plane flying?

A. 3hrs B. 6hr C. 12hr D. 10hr

25. The solution set of f(x) = $\frac{x-2}{x^2-4} < o$ *is* _____

A. $(-\infty, -2)U(1, \infty)$ B. $(-\infty, -2)U[1, \infty)$ C. $(-\infty, -2)$ D. $(-\infty - 2)U(2, \infty)$

26. Which one of the following is true about f(x) = $\frac{x^2-4}{x+2}$?

A. vertical asymptote x= -2 C. The graph of the function has a hole at x= -2

B. horizontal asymptote x= ± 2 D. *oblique asymptote* $y = x$

II. Short answer (Write the answer only.

27. Find the area of the triangle with vertices the following three set of point.

(1,-1), (1,2) and (3,2)

28. Given f(x)= 2-2 sgnx, then find the solution set of $|f(x)|$

29. Let A= $\begin{pmatrix} a+b & 8 \\ 5 & -1 \end{pmatrix}$ and B = $\begin{pmatrix} 8 & 8 \\ a-b & -1 \end{pmatrix}$. If A=B then determine the value of a and b.

30. For what value of x is A= $\begin{pmatrix} 1 & x & 0 \\ 0 & 2 & 0 \\ 3 & 4 & 3 \end{pmatrix}$ is lower triangle matrix.

31. The perimeter of a triangle is 50cm. The longest side of the triangle measures 20cm more than the shortest side. The middle side is 2 times the measure of the shortest side. Find the length of the three sides of the triangle.

32. Let A= $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ then find the determinate of 4A

Blessed Cr Michael Catholic Secondary School

2st Semester Mathematics pre-test one for Grade 12th

Name Hanna Ayalew NO 24 Section 12th C

1. A
2. B
3. B
4. A
5. C
6. A
7. C
8. D
9. A
10. C
11. D
12. B
13. A
14. B
15. A
16. D
17. C
18. B
19. C
20. B
21. C
22. D

I. Short answer

23. B
24. A
25. C
26. C
27. 1
28. 0
29. $a=\frac{13}{2}$ $b=\frac{3}{2}$
30. $X=0$
31. $(x, y, z) = (27.5, 15, 7.5)$
32. -32

30

Excellent

Blessed G/Michael Catholic General Secondary School
Mathematics pre- test one for grade 12th
Time allowed 1:30

I. Choose the best answer among the given alternatives.

-----1. If $x = 0.\dot{3}$ and $y = 2.3\dot{2}$, then the value of x+y is ----------

A. $\frac{239}{90}$ B. $\frac{249}{90}$ C. $\frac{259}{90}$ D. $\frac{340}{90}$

-----2. Let x = natural number and y = Irrational number then which one of the following is necessarily true?

A. $\frac{y}{x}$ is real number
B. xy is an irrational number
C. x+y is an irrational
D. $\frac{x}{y}$ is an irrational

-----3. Which one of the following is true?

A. a > b, b > 0, a < b, then $\sqrt{a} > \sqrt{b}$
B. $\frac{22}{7}$ is a rational number
C. π is rational number
D. a > 0, b>0, then $\sqrt{\frac{a}{b}} \neq \frac{\sqrt{a}}{\sqrt{b}}$

-----4. The simplification form of $\sqrt{45} \times \sqrt[3]{\sqrt{125}}$ = ----------

A. 15 B. 25 C. 30 D. 4

-----5. Let x – be real number, then which one of the following statement is not true.

A. If a > 0, then $a > \sqrt{a}$
B. If a > 1, then $\sqrt{a} > \sqrt[3]{a}$
C. If a > 1, then $a > \sqrt{a}$
D. If 0 < a < 1 then $\sqrt{a} > \sqrt[3]{a}$

----6. If both x and y are rational numbers, then the value of x and y If $\frac{\sqrt{3}+\sqrt{2}}{3\sqrt{2}-2\sqrt{3}} = x - y\sqrt{6}$:

A. $x = 2$, $y = 5\sqrt{6}$ B. $x = 2$, $y = -5\sqrt{6}$ C. $x = 5\sqrt{6}$, $y = 2$ D. $x = \sqrt{6}$, $y = -2$

------7. $\frac{2\sqrt{3}}{\sqrt{3}+\sqrt{2}} - \frac{\sqrt{3}}{\sqrt{3}-\sqrt{2}}$ is equal to :-

A. $3 - 3\sqrt{6}$ B. $2\sqrt{3}$ C. $\sqrt{3}$ D. $3+2\sqrt{3}$

-----8. $\left(\frac{1}{-512}\right)^{1/3}$ is equal to

A. $^1/_4$ B. $^{-1}/_4$ C. $\frac{1}{8}$ D. $\frac{-1}{8}$

-----9. $\left(\frac{25}{9}\right)^{1-x}\left(\frac{125}{27}\right)^{x} = \left(3/5\right)^{x+2}$, the value of x is ----------

A. 4 B. -2 C. -4 D. 1

----10. The value of a and b are constant for which (3, -1) is a solution of the system $\begin{cases} ax + by = 16 \\ bx - 3y = 0 \end{cases}$ then the value of a is equal to - - -

A. -1 B. -4 C. 5 D. 6

-----11. The solution set of $\begin{cases} -\frac{1}{2}y + 3x = 6 \\ -2x + y = 2y + 4 \end{cases}$ is equal to

A. $\{(0,2)\}$ B. $\{(1,-6)\}$ C. $\{\ \}$ D. $\{(-6,4)\}$

-----12. If the system $\begin{cases} ay + 3x = 7 \\ 5y = -2x + b \end{cases}$ has one solution then the value of a and b is equal to:------

A. $A \neq 15/2,\ b \in IR$ C. $a = \frac{15}{2}, b \in IR$

B. $a = 15/2, b \in IR$ D. $a \neq 15/2$ and $b = 7$ only

-----13. The graph of $y = -2x^2 - bx - c = 0$ passing through (-2, -3) and (1, 15), then the value of b and c is ----------

A. (8, 5) B. (1, 5) C. (2, 6) D. (4,5)

-----14. The solution set of $|3 - |x + 4|| - 8 = 0$ is ----------

A. 7 and -15 B. -7 and 15 C. 3 and 4 D. 1 and 15

-----15. If one of the roots of the equation $x^2 - 8x = -k$ exceeds the other by 4, then the value of k is ----------.

A. 4 B. 2 C. 12 D. 6

-----16. Let A= {1, {2,3}} and B= {1,2,3}, then which one of the following is true.

A. A/B = {{1,2,3}} C. AnB = {1,2,3}

B. B/A = {2,3} D. AuB = {1,2,3,{2}}

-----17. If AnB = {-1,0,1} and BnC = {2,-1,0}, then Bn (Auc) is equal to -------

A. {0, -1} B. {-1,0, 1} C. {0, -1,2} D. {-1,0,1,2}

-----18. Which one of the following is not a subset of the set {0, Ø}.

A. Ø B. { 0} C. 0 D. {Ø}

-----19. If A and B are two sets such that n(A) =6, n (B) =4 and n(AnB)=2, then n(AΔB) is equal to:

A. 6 B. 4 C. 8 D. 2

-----20. If n(u) = 30, n(A) =6, n (B) = 10, n(AuB) = 14, then the n(A') is ----------,

A. 16 B. 24 C. 30 D. 36

-----21. Which one of the following is true about the relation R ={(x,y): y =$\sqrt{x}$?

A. Domain R =IR +

B. Range of R =[0,∞)

C. (-4,2)∈ IR

D. domain of R = [1,∞)

-----22. Let R = {(x,y) : y≥ x^2-4 and y ≤ 5, then the ~~range~~ domain of R is:

A. [-3,3] B. (0, 3) C. (0, ∞) D. [-3, ∞)

-----23. If f(x) = x and g(x) = x+1, the value of $(2f+\frac{1}{2}g)(1)$ = _____

A. 3 B. 1 C. 2 D. 4

-----24. Which one of the following is not true about f(x) = $2x^2$-4x+3:

A. The graph opens upward

B. The domain is all real number

C. range is y ≥ 0

D. vertex is (2,3)

-----25. Which of the following is true about f(x) = x^2+4x+2

A. Vertex (-2, -2)

B. down ward parabola

C. domain is (-2,∞)

D. range IR.

Blessed G/ Michael Catholic Secondary school Mathmatics pretest one exam for Grade 12 students

OCTOBER/2025 **Time allowed- 1 hr:10 min**

Name Hanna Ayalew **Sec.** 12th C **R.No.** 23

27

1. A
2. C
3. B
4. A
5. D
6. ~
7. A
8. D
9. B
10. C
11. B
12. A
13. A
14. A
15. C
16. B
17. D
18. C
19. C
20. B
21. B
22. B
23. A
24. C
25. A
26. ----------
27. ----------
28. ----------
29. ----------
30. ----------
31. ----------
32. ----------
33. ----------
34. ----------
35. ----------
36. ----------
37. ----------
38. ----------
39. ----------
40. ----------

Blessed G/Michael Catholic General Secondary School
Mathematics pre- test two for grade 12th

Time allowed 1:40

I. Choose the best answer among the given alternatives.

----1. Let A = {1, 0 ,2} and B= {-1, 2, 4}, then which one of the following is true about the relation A to B, for R = { (x,y) : y = x -1.

A. elements of R = {(1,0), (0,-1), (2,1)}.
B. domain R= {-1,2,4}
C. Range of R = {1,0,2}
D. elements of R = {(1,-1), (0,2), (2,4)}

----2. If A= {0,1,2} and B= {0,3,4} and R is a relation from B to A such that R= {(x,y): $y=\sqrt{x-1}$, then what is the range of R:

A. IR B. Not exist C. {1,2} D. {0,1, 2}.

----3. Which one of the following is not true?

A. Every function is a relation
B. The range R= {(x,y) :y = /x/, its range for x∈ IR is [0,∞)
C. The range of the constant function is IR
D. Domain is the set of the first element of the ordered pair

----4. Let R = {(x,y): y≤ 4 - x, y≤ x+1 and y≥-1, then which of the following is true about R.

A. Domain of R = [-2,5].
B. Range of R = [-1, 5]
C. The set of the R is about x-axis
D. domain R^{-1} = [-1, 5]

----5. Let R = {(x,y) :y+4 ≥x^2 and 5≥ y, the which one of the following is not true.

A. Domain of R = [-3,3)
B. Range of R = [-4,5]
C. domain of R^{-1} = [-4, 5]
D. (-1,-3) ∈ R

----6. Which one of the following Relation is a function?

A. R = {(x,y): x =y^2}
B. R = {(x,y) : x =3}
C. R = {(x,y) : x = $(y+3)^2$}
D. R = {(x,y) : y = x+2}

-----7. If f(x) = x^3+Ax^2 + 4x-5 and f(-1) = 10, what is the value of A?

A. 20 B. -40 C. -20 D. 40

-----8. Let f(x) = x-4 and g(x) = $\sqrt{3-x}$, then the domain of (f+g) (x) is ----------

A. IR B. (-∞, 3) C. (-∞,3] D. -3≤ x<3

-----9. If f = {(1,2), (2,3), (3,5)}and g(x) = $\frac{1}{2}x + \frac{3}{2}x^2$, then g/h(3)

A. 15 B. cannot determine C. 3 D. 5 ∞

-----10. Let $f(x)=\begin{cases} x+2 \text{ if } x \le -1 \\ x^3 \text{ if } -1 < x \le 1 \\ 3-x \text{ if } x \ge 1 \end{cases}$ and $g(x):\begin{cases} x-3 \text{ if } x \le -2 \\ -x^2 \text{ if } -2 < x \le 3 \\ 4-x \text{ if } x \ge 3 \end{cases}$

then $(f+g)(-3/2)$ is ----------.

A. 7/4 B. -7/4 C. 4/3 D. -4/3

------11. Let f(x) = (-a+1) x+2-b, where a and b are positive real numbers, which of the following is true about the graph.

A. Its slope is – a

B. It is increasing for a < 1

C. The y – intercept is 2

D. The x – intercept is $\frac{b-2}{a-1}$

------12. Let $f(x) = x^3+2x^2- 3ax -10$ divided by x+3 has a remainder of 8, the value of a is: ------

A. 4 B. 7 C. 3 D. 1

------13. $f(x) = Ax^2-Ax+Bx +A-2B$ where A and B are constant when f(x) is divided by (x+3) and (x-3) the remainder are 58 and 10 respectively. Which one of the following is the value of A and B.

A. 3, -2 B. 4, 1 C. 2, 4 D. 1, 2

------14. If the polynomial $f(x) = x^3+2x+2$ and $g(x)= x^3+3x-4$ have the same remainder when they have divided by (x-k), Then the value of k is ----------

A. 4 B. 6 C. 2 D. 1

------15. Which one of the following is the value of A and B so that the polynomial $x^3- Ax^2-13x+B$ has x-1 and x-3 as a factor is:

A. (3,10) B. (3,2) C. (3,15) D. (15,2)

------16. A polynomial function of degree 3 such than f(1)= 36 and x-2, x+2 and x+1 are factors of the polynomial, then which one is represent a function of f.

A. $f(x) = -6x^3+6x^2-4x$

B. $f(x) = -6x^3-6x^2+24$

C. $f(x) = -6x^3-6x^2+24x+24$

D. $f(x) = 6x^3-6x^2-24x+6$

------17. If f is a polynomial function given by $f(x) = 2x^3-4x^2+5x+4$ then this function has root between ----------.

A. 1 and 3 B. -1 and 1 C. 0 and 2 D. 1 and 2

------18. Which one of the following function touches but never crosses the x- axis.

A. $f(x) = 1-x^3$ B. f(x) = x4 -16 C. $f(x) = x^4-2x^2+1$ D. $f(x) = x(1-x^4)$

------19. Which of the following is true about $f(x) = x^5(x-1)^2 (x^2+1)$

A. 0 is a root of f(x) crosses the x – axis at x =1

B. The graph of f(x) crosses the x- axis at x = -1

C. The graph of f(x) crosses the x – axis at x = 1

D. f(x) has three distnic real root.

-----20. If $f(x) = 2x^3-kx^2+x-2k$, then a number k such that the graph of f contains the point (1,6) is ------

A. 4 B. -1 C. 2 D. -2

-----21. Which one of the following is true?

A. $\left(\frac{1}{2}\right)^x > (2/3)^x$, for $x > 0$ C. $\left(\frac{1}{2}\right)^x < 0$, for $x < 0$

B. $\left(\frac{1}{2}\right)^x < (2/3)^x$ for $x < 0$ D. The graph of $y = \left(\frac{1}{3}\right)^x$ decreasing from $x \to \infty$

------22. The solution set of $(3)^{-2-x}(9)^{2x} = \frac{27}{9x}$ is ----------

A. {4} B. {1} C. {-1} D. {0}

-----23. Which one of the following is the value of x for $10^x+(10)^{x-1} = \frac{11}{100}$?

A. 3 B. 2 C. -1 D. -2

------24. If $(64)^y = \left(\frac{1}{8}\right)^x$, then $\frac{x}{y}$ = ----------.

A. -4 B. 4 C. 2 D. -6

------25. The simplified form of $-\log_2^{8}\sqrt[3]{2}$ is ----------.

A. $\frac{10}{3}$ B. $\frac{3}{10}$ C. $\frac{-3}{10}$ D. $\frac{-10}{3}$

------26. $(x)^{\log^{y}\sqrt[3]{x}}$ is equal to: ----------

A. $\frac{y}{3}$ B. 3y C. y D. y^3

------27. If $x = \log^{4} k$, $y = \log^{3}_{k}$ and $\log^{b}_{k} = z$, then $\log^{36}_{b}$ is ----------.

A. $\frac{y+x}{z}$ B. $\frac{y^2+x}{2}$ C. $\frac{x+2y}{z}$ D. $\frac{x-y}{z}$

-------28. If $\log^{4}_{a} = x$, then $\log\frac{16}{\sqrt{2a}}$ is equal to : ----------

A. $\frac{8x}{x+2}$ B. $\frac{x}{x+2}$ C. $\frac{4x}{x-2}$ D. $\frac{x}{2x+1}$

-------29. If $a > 1$, then for what value of x is $\log^{x}_{a} < 0$?

A. $(1, \infty)$ B. $(0,\infty)$ C. $(-\infty, 0)$ D. (0, 1)

------30. Which one of the following is the value of x for $\log^{(x-4)} = log^{\frac{1}{x}} + \log(3x-10)$

A. 2 B. 4 C. 5 D. 1

25

Mathematics Answer Sheet For Grade 12.

Name Hanna Ayalew **Sec.** 12TH C **No.** 24 **Time allotted 1.40'**

1. A
2. D
3. C
4. A
5. A
6. D
7. A
8. C
9. C
10. B
11. A
12. C
13. A
14. A
15. C
16. C
17. B
18. C
19. C
20. B
21. D
22. B
23. C
24. C 1-21
25. D
26. D
27. B
28. C
29. D
30. C